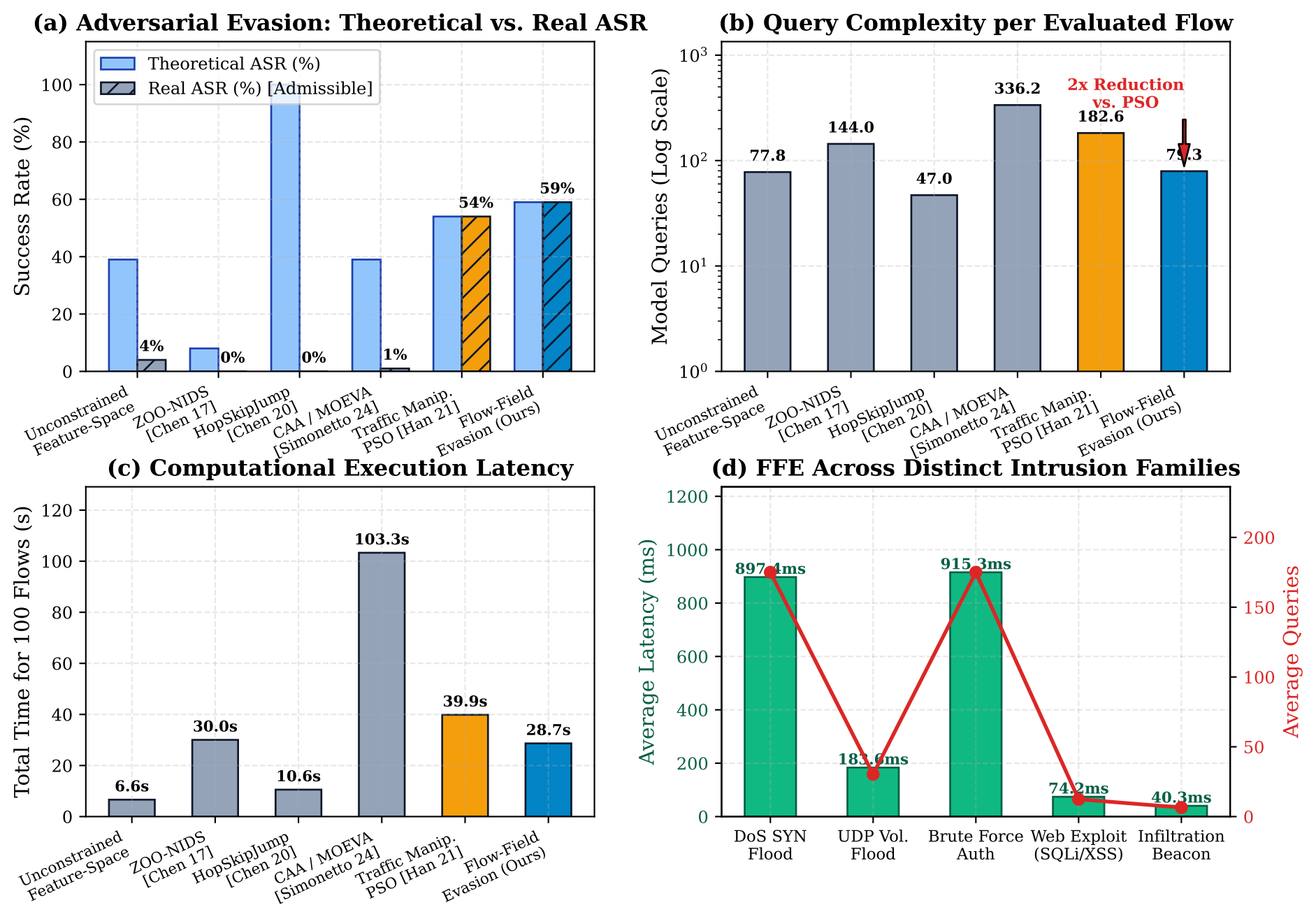
(a) Adversarial Evasion: Theoretical vs. Real ASR
Theoretical ASR (%)
Real ASR (%) [Admissible]
Success Rate (%)
4%
0%
0%
1%
54%
59%
Unconstrained Feature-Space
ZOO-NIDS [Chen 17]
HopSkipJump [Chen 20]
CAA / MOEVA [Simonetto 24]
Traffic Manip. PSO [Han 21]
Flow-Field Evasion (Ours)
(b) Query Complexity per Evaluated Flow
Model Queries (Log Scale)
77.8
144.0
47.0
336.2
182.6
79.3
2x Reduction vs. PSO
(c) Computational Execution Latency
Total Time for 100 Flows (s)
6.6s
30.0s
10.6s
103.3s
39.9s
28.7s
(d) FFE Across Distinct Intrusion Families
Average Latency (ms)
Average Queries
897.4ms
183.6ms
915.3ms
74.2ms
40.3ms
DoS SYN Flood
UDP Vol. Flood
Brute Force Auth
Web Exploit (SQLi/XSS)
Infiltration Beacon